PANH: A PADÉ APPROXIMATED NON-LINEAR HYPERBOLIC ACTIVATION FUNCTION FOR PHYSICS-INFORMED NEURAL NETWORKS

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Abstract. Physics-informed neural networks frequently suffer from optimization difficulties and vanishing gradients when relying on standard saturating activation functions. To address these limitations, we introduce an adaptive rational activation function grounded in theoretical approximation principles that guarantees non-vanishing gradients while maintaining extreme smoothness. Empirical evaluations across diverse differential equation benchmarks demonstrate that this approach autonomously adapts to various functional regimes, significantly improving convergence and accuracy, particularly for high-frequency and multi-scale problems.

Keywords: Physics-Informed Neural Networks, Adaptive Activation Functions, Padé Approximation, Vanishing Gradients, Spectral Bias, Scientific Machine Learning.

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1. Introduction

The choice of activation function is a highly consequential design decision in deep learning architectures. It governs the nonlinear expressivity of each layer, the topology of the optimization landscape, and, critically in the context of scientific computing, the well-posedness of high-order automatic differentiation. Classical choices such as the hyperbolic tangent (tanh) offer smooth and bounded outputs, but their saturating behavior for large pre-activations introduces the vanishing gradient problem [1]: the derivative $\tanh'(x) = \operatorname{sech}^2(x)$ decays exponentially to zero, effectively halting the gradient signal in deep layers.

This pathology is severely exacerbated in Physics-Informed Neural Networks (PINNs) [4], where the model must approximate both the state variable and its high-order spatial and temporal derivatives. In such architectures, the activation must be at least C^2 (and practically C^∞) to ensure that automatic differentiation of the PDE residual is numerically stable. Furthermore, standard saturating networks exhibit a well-documented spectral bias, inherently struggling to capture high-frequency components [7]. While periodic activations such as SIREN [6] mitigate spectral bias, they require highly specialized initialization schemes and often introduce unstable oscillatory artifacts. Any gradient attenuation compounded with PDE stiffness produces severe optimization bottlenecks that standard learning rate schedules cannot fully resolve.

Motivated by these constraints, recent literature has explored *adaptive* activations, wherein the functional shape is parameterized by learnable coefficients optimized concurrently with the network weights [5, 2]. The underlying premise is that distinct physical regimes—saturating, oscillatory, convective—demand fundamentally different nonlinear profiles, and that fixing the activation *a priori* systematically disadvantages certain problem classes. However, existing adaptive designs generally suffer from two critical shortcomings: (i) the initialization of their free parameters is either arbitrary or heuristic, risking unbounded forward passes early in training; and (ii) no theoretical connection exists to classical, well-understood functions, making convergence guarantees elusive.

This work addresses both shortcomings by introducing the **Panh** (*Padé Approximated Non-linear Hyperbolic*) activation function. Panh is a rational function of degree $(3, 2)$ governed by two learnable scalar parameters, α and β. Its design rests on two complementary pillars: exact grounding in Padé approximation theory for mathematically sound initialization, and strict non-saturating asymptotic properties for robust gradient flow.

Beyond these pillars, Panh exhibits a rich analytical structure. It presents a three-inflection-point geometry, a provably factored second derivative, an identity-collapse condition with a clear algebraic interpretation, and a full parameterization of the convexity-concavity partition. All of these properties are derived in closed form in Section 2.

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The remainder of this paper is organized as follows. Section 2 establishes the complete mathematical analysis of Panh. Section 3 studies the learning dynamics of the adaptive parameters. Section 4 presents comprehensive numerical benchmarks on PINNs problems. Finally, Section 5 outlines concluding remarks.

2. Mathematical Analysis of the Panh Activation

2.1. **Definition and fundamental properties.**

Definition 2.1 (Panh activation function)**.** Let $\alpha, \beta \in \mathbb{R}$ be learnable scalar parameters. The *Panh* activation function is defined by

$$f_{\alpha,\beta}\colon \mathbb{R} \longrightarrow \mathbb{R}, \qquad f_{\alpha,\beta}(x) = \frac{x+\alpha x^3}{1+|\beta|\,x^2} = \frac{x\,(1+\alpha x^2)}{1+|\beta|\,x^2}. \tag{1}$$

The key structural decision in (1) is the use of $|\beta|$ rather than β in the denominator. Setting $b := |\beta| \geq 0$, one guarantees $1 + bx^2 \geq 1$ for all $x \in \mathbb{R}$, ensuring the denominator is strictly positive regardless of the sign of the raw parameter β. This formulation allows unconstrained optimization via gradient-based methods while preserving two foundational properties.

Proposition 2.2 (Domain and regularity)**.** *For any $\alpha, \beta \in \mathbb{R}$, we have* $\mathrm{Dom}(f_{\alpha,\beta}) = \mathbb{R}$ *and* $f_{\alpha,\beta} \in C^\infty(\mathbb{R})$.

Proof. Since $b = |\beta| \geq 0$, the denominator satisfies $1 + bx^2 \geq 1 > 0$ for all $x \in \mathbb{R}$. The function $f_{\alpha,\beta}$ is therefore a ratio of two polynomials with a globally positive denominator, which is infinitely differentiable on $\mathbb{R}$. □

Proposition 2.3 (Odd symmetry)**.** *The Panh function is odd: $f_{\alpha,\beta}(-x) = -f_{\alpha,\beta}(x)$ for all $x \in \mathbb{R}$.*

Proof. Direct substitution gives $f_{\alpha,\beta}(-x) = (-x)(1+\alpha x^2)/(1+bx^2) = -f_{\alpha,\beta}(x)$. □

Remark 2.4. Odd symmetry implies that the activation is zero-centered: $\mathbb{E}[f_{\alpha,\beta}(Z)] = 0$ for any zero-mean symmetric random variable Z. This property is empirically known to improve optimization dynamics by preventing systematic bias accumulation across layers [1].

Proposition 2.5 (Local identity at the origin)**.** $f_{\alpha,\beta}(0) = 0$ *and* $f'_{\alpha,\beta}(0) = 1$.

The condition $f'_{\alpha,\beta}(0) = 1$ ensures that Panh acts locally as the identity map near the origin. This is consistent with unit-Jacobian initialization strategies and prevents output scale distortion common to activations like ReLU or tanh at initialization.

2.2. **Padé connection and initialization.**

The rational structure of Panh naturally aligns with the family of Padé approximants, which provide the optimal rational approximation to a smooth function via Taylor coefficient matching [3]. The following result demonstrates that Panh, at a specific parameter pair, is precisely the Padé $[3/2]$ approximant of tanh.

Theorem 2.6 (Padé–tanh equivalence)**.** *Set $\alpha_0 = 1/15$ and $b_0 = |\beta_0| = 2/5$. Then*

$$f_{\alpha_0,\beta_0}(x) = \tanh(x) + \mathcal{O}\big(x^7\big)\,, \qquad x \to 0. \tag{2}$$

That is, f_{α_0,β_0} is the Padé $[3/2]$ approximant of tanh *centered at the origin.*

Proof. Using the geometric expansion $(1+bx^2)^{-1} = \sum_{k\geq 0}(-b)^k x^{2k}$ valid for $\left|bx^2\right| < 1$, we compute:

$$f_{\alpha,\beta}(x) = (x+\alpha x^3)\sum_{k=0}^{\infty}(-b)^k x^{2k} = x + (\alpha-b)x^3 + (b^2-\alpha b)x^5 + \mathcal{O}\big(x^7\big)\,. \tag{3}$$

The Maclaurin series of tanh reads:

$$\tanh(x) = x - \tfrac{1}{3}x^3 + \tfrac{2}{15}x^5 - \tfrac{17}{315}x^7 + \mathcal{O}\big(x^9\big)\,. \tag{4}$$

Matching the coefficients of x^3 and x^5 in (3) with those of (4):

$$\begin{aligned} x^3&: \quad \alpha - b = -\tfrac{1}{3} \implies b = \alpha + \tfrac{1}{3}, \\ x^5&: \quad b(b-\alpha) = \tfrac{2}{15} \implies b\cdot\tfrac{1}{3} = \tfrac{2}{15} \implies b = \tfrac{2}{5}. \end{aligned}$$

Back-substituting gives $\alpha = 2/5 - 1/3 = 1/15$. With these values, the coefficients of x^3 and x^5 coincide precisely with those of tanh, relegating the leading error to $\mathcal{O}\big(x^7\big)$. □

Remark 2.7 (Initialization strategy)**.** Theorem 2.6 establishes a rigorous initialization strategy: setting $(\alpha, \beta) = (1/15, 2/5)$ ensures the network behaves exactly as a standard tanh model on any bounded pre-activation range. As gradients backpropagate, parameters α and β adapt securely to the geometry of the specific loss landscape.

The fidelity of this approximation is highlighted in Figure 1: the absolute error remains below 10^{-4} throughout the standard pre-activation range $[-2, 2]$.

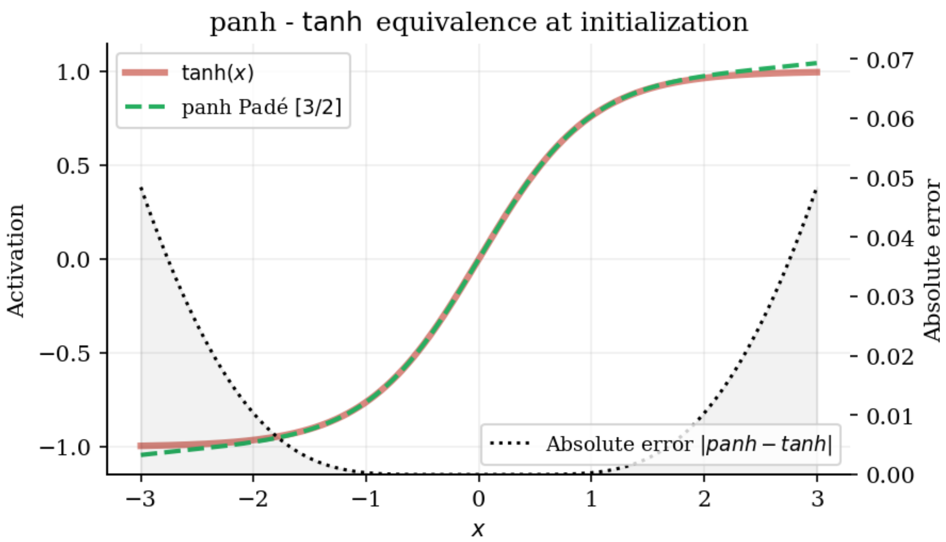

FIGURE 1. Equivalence between Panh at the Padé initialization $(\alpha_0, b_0) = (1/15, 2/5)$ and tanh. The absolute error (lower panel, log scale) confirms agreement up to $\mathcal{O}(x^7)$ near the origin.

2.3. **Parametric family and functional regimes.** A defining characteristic of Panh is its capacity to model diverse functional shapes. By varying α and $|\beta|$, the network accesses qualitatively distinct nonlinear regimes.

Proposition 2.8 (Functional regime classification)**.** *Let $b = |\beta| > 0$. The asymptotic behavior of $f_{\alpha,\beta}$ is governed by α:*

(I) ***Non-saturating regime*** *($\alpha > 0$): $f_{\alpha,\beta}(x) \sim (\alpha/b)\,x$ as $|x| \to \infty$. The function is unbounded and strictly increasing when $b < 3\alpha$ or when the numerator polynomial has no real root (see Section 2.5).*

(II) ***Bounded-bump regime*** *($\alpha = 0$): $f_{\alpha,\beta}(x) = x/(1 + bx^2) \to 0$ as $|x| \to \infty$. The function is bounded, achieving a global maximum of $1/(2\sqrt{b}\,)$ at $x = 1/\sqrt{b}$.*

(III) ***Non-monotone regime*** *($\alpha < 0$): $f_{\alpha,\beta}(x) \to -\operatorname{sign}(x) \cdot \infty$ as $|x| \to \infty$. The function possesses local extrema dependent on α and b.*

Proof. Regimes (I) and (III) follow immediately from dividing numerator and denominator by x^2: $f_{\alpha,\beta}(x)/x \to \alpha/b$. Regime (II) reduces to $f_{\alpha,\beta}(x) = x/(1 + bx^2)$. Differentiation gives $f'_{\alpha,\beta}(x) = (1 - bx^2)/(1 + bx^2)^2$, which vanishes at $x = \pm 1/\sqrt{b}$, confirming the location of the extrema. □

The three regimes are illustrated in Figure 2. The transition between regimes occurs continuously as α crosses zero, providing the optimizer with a smoothly explorable parameter space.

2.4. **Gradient analysis and vanishing gradient prevention.**

Proposition 2.9 (Closed-form first derivative)**.**

$$f'_{\alpha,\beta}(x) = \frac{1 + (3\alpha - b)\,x^2 + \alpha b\,x^4}{(1 + bx^2)^2}, \qquad b = |\beta|\,. \tag{5}$$

Proof. Setting $N(x) = x + \alpha x^3$ and $D(x) = 1 + bx^2$, the quotient rule yields:

$$f'_{\alpha,\beta}(x) = \frac{N'D - ND'}{D^2} = \frac{(1 + 3\alpha x^2)(1 + bx^2) - (x + \alpha x^3)(2bx)}{(1 + bx^2)^2}.$$

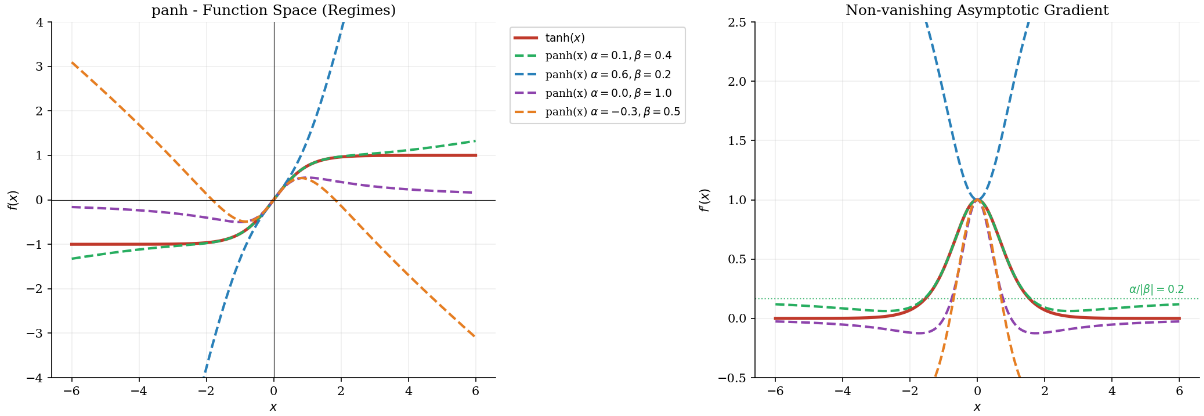

FIGURE 2. **Left**: The Panh parametric family displaying three qualitative regimes. **Right**: Corresponding gradients. While $\tanh'$ vanishes exponentially, each Panh variant maintains a finite asymptotic plateau ($\alpha/|\beta|$), preventing gradient saturation.

Expanding the numerator: $(1+3\alpha x^2)(1+bx^2)-2bx^2(1+\alpha x^2) = 1+(3\alpha-b)x^2+\alpha bx^4$, which establishes (5). □

Theorem 2.10 (Non-vanishing asymptotic gradient)**.** *For $b = |\beta| > 0$,*

$$\lim_{|x|\to\infty} f'_{\alpha,\beta}(x) = \frac{\alpha}{|\beta|}. \tag{6}$$

In particular, when $\alpha > 0$, the gradient is bounded strictly away from zero over $\mathbb{R}$.

Proof. Divide numerator and denominator of (5) by x^4:

$$f'_{\alpha,\beta}(x) = \frac{x^{-4} + (3\alpha - b)x^{-2} + \alpha b}{b^2 + 2bx^{-2} + x^{-4}} \xrightarrow{|x|\to\infty} \frac{\alpha b}{b^2} = \frac{\alpha}{b}.$$

□

Remark 2.11. Theorem 2.10 articulates the central advantage of Panh over standard saturating functions. As backpropagation recursively multiplies activation derivatives across L layers, the exponential decay of $\tanh'$ induces catastrophic signal attenuation. Conversely, Panh (with $\alpha > 0$) strictly bounds each factor by $\alpha/b > 0$, ensuring robust gradient flow through arbitrary network depths.

2.5. **Monotonicity.** Let $P(x) := 1 + (3\alpha - b)x^2 + \alpha b\, x^4$ denote the numerator of $f'_{\alpha,\beta}$. Viewed as a polynomial in $u = x^2 \geq 0$, $P(u) = \alpha b\, u^2 + (3\alpha - b)\, u + 1$.

Theorem 2.12 (Global monotonicity)**.** *If $\alpha > 0$ and the discriminant $\Delta := (3\alpha - b)^2 - 4\alpha b = 9\alpha^2 - 10\alpha b + b^2$ is non-positive, then $f'_{\alpha,\beta}(x) > 0$ for all $x \in \mathbb{R}$, and hence $f_{\alpha,\beta}$ is strictly increasing.*

The Padé initialization $(\alpha_0, b_0) = (1/15, 2/5)$ satisfies $\Delta = 9/225 - 10/75 + 4/25 = -5/225 < 0$, ensuring Panh is strictly monotone at the start of training.

Proof. Since $(1 + bx^2)^2 > 0$, the sign of $f'_{\alpha,\beta}$ equals the sign of P. As a quadratic in $u = x^2 \geq 0$, P possesses no positive real roots if and only if either $\Delta \leq 0$ or all coefficients share the same sign. When $\Delta < 0$ and the leading coefficient $\alpha b > 0$, $P(u) > 0$ for all $u \in \mathbb{R}$, thus naturally over $u \geq 0$. □

2.6. **Second derivative and inflection structure.** The second derivative dictates the local curvature of the loss surface with respect to network inputs. For Panh, it admits a remarkably compact factorization.

Theorem 2.13 (Factored second derivative)**.**

$$f''_{\alpha,\beta}(x) = \frac{2x\,(\alpha - |\beta|)\,(3 - |\beta|\; x^2)}{(1 + |\beta|\; x^2)^3}. \tag{7}$$

Proof. Differentiating (5) utilizing the quotient rule with $P = 1 + (3\alpha - b)x^2 + \alpha bx^4$ and $Q = (1 + bx^2)^2$:

$$P' = 2(3\alpha - b)x + 4\alpha bx^3 = 2x\big[(3\alpha - b) + 2\alpha bx^2\big],$$
$$Q' = 4bx(1 + bx^2).$$

Factoring $2x(1+bx^2)$ from the numerator of $f''_{\alpha,\beta} = (P'Q - PQ')/Q^2$ yields:

$$\begin{aligned}&\big[(3\alpha - b) + 2\alpha b x^2\big](1+bx^2) - 2b\big[1 + (3\alpha - b)x^2 + \alpha b x^4\big]\\ &= (3\alpha - b) + b(3\alpha - b)x^2 + 2\alpha b x^2 + 2\alpha b^2 x^4 - 2b - 2b(3\alpha - b)x^2 - 2\alpha b^2 x^4\\ &= 3(\alpha - b) - b(\alpha - b)x^2 = (\alpha - b)(3 - bx^2).\end{aligned}$$

Dividing by $(1+bx^2)^3$ generates (7). □

Corollary 2.14 (Three inflection points)**.** *Assume $b = |\beta| > 0$ and $\alpha \neq b$. Then $f''_{\alpha,\beta}(x) = 0$ at exactly three points:*

$$x = 0, \qquad x = \pm x^* := \pm\sqrt{\frac{3}{|\beta|}}. \tag{8}$$

Proof. From (7): $f''_{\alpha,\beta}(x) = 0 \iff x = 0$ or $bx^2 = 3$, since $\alpha \neq b$ prevents $(\alpha - b)$ from vanishing. □

This contrasts with tanh, which possesses a single inflection point at the origin. Panh provides two additional lateral inflection points, placed symmetrically at $\pm x^*$. At Padé initialization, $x^* = \sqrt{3/(2/5)} \approx 2.74$. These lie outside the standard active pre-activation range $[-2, 2]$, ensuring Panh behaves akin to tanh centrally while offering superior geometric flexibility for extreme activations.

Proposition 2.15 (Convexity partition)**.** *Assume $\alpha > b > 0$. The convexity structure of $f_{\alpha,\beta}$ is partitioned as follows:*

Interval	sign(x)	sign$(3 - bx^2)$	sign$(f''_{\alpha,\beta})$
$(-\infty, -x^*)$	$-$	$-$	$+$ *(convex)*
$(-x^*, 0)$	$-$	$+$	$-$ *(concave)*
$(0, x^*)$	$+$	$+$	$+$ *(convex)*
$(x^*, +\infty)$	$+$	$-$	$-$ *(concave)*

The entire structure reverses when $\alpha < b$.

This alternating curvature pattern, visualized in Figure 3, holds significant computational implications for PINNs. It assists the network in representing Laplacian sign changes natively within the activation function, reducing required network depth.

2.7. **Fixed points and the identity-collapse condition.**

Proposition 2.16 (Fixed points)**.** *The equation $f_{\alpha,\beta}(x) = x$ holds if and only if $x = 0$, or $\alpha = |\beta|$.*

Proof. $f_{\alpha,\beta}(x) = x \iff x(1+\alpha x^2) = x(1+bx^2) \iff x(\alpha - b)x^2 = 0$. This mandates $x = 0$ or $\alpha = b$. □

When $\alpha = |\beta|$, the activation degenerates strictly to the identity map $f_{\alpha,\beta}(x) = x$:

$$\alpha = |\beta| \implies f_{\alpha,\beta}(x) = \frac{x(1+\alpha x^2)}{1+\alpha x^2} = x. \tag{9}$$

This induces *identity collapse*: the layer contributes no nonlinearity. While occasionally beneficial for redundant individual neurons, systemic collapse eliminates the network's expressive capacity. Consequently, we define the *collapse monitor*:

$$\delta(t) := |\alpha(t) - |\beta(t)|| \tag{10}$$

Tracking $\delta(t)$ during training serves as a diagnostic indicator of topological degeneracy.

3. Learning Dynamics and Parameter Evolution

3.1. **Loss landscape and the role of the Padé initialization.** Figure 4 visualizes the empirical Mean Squared Error surface as a function of $(\alpha, |\beta|)$. The Padé initialization $(\alpha_0, b_0) = (1/15, 2/5)$ resides securely within a broad, optimal basin. Empirical gradient descent trajectories originating from varied initial states all systematically converge into this optimal valley, verifying the geometric stability of the initialization.

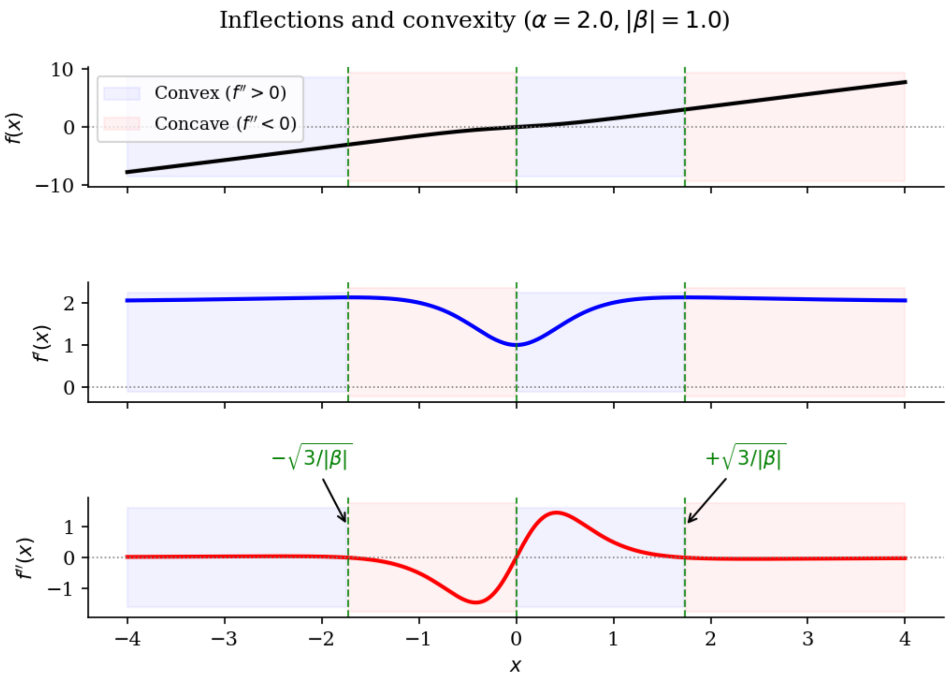

FIGURE 3. Activation f, first derivative f', and second derivative f'' at Padé initialization. The three inflection points are annotated, explicitly defining the convex and concave regions.

TABLE 1. Comparative summary of properties between Panh ($\alpha > 0$) and tanh.

Property	Panh ($\alpha > 0$)	tanh
Domain	$\mathbb{R}$	$\mathbb{R}$
Regularity	$C^\infty(\mathbb{R})$	$C^\infty(\mathbb{R})$
Odd symmetry	✓	✓
$f(0) = 0,\ \ f'(0) = 1$	✓	✓
Bounded	No	Yes
Asymptotic gradient	$\alpha/\,\lvert\beta\rvert > 0$	0
Number of inflection points	3	1
Vanishing gradient problem	No	Yes
Learnable shape parameters	2 (α, β)	0
Coincides with tanh (up to $\mathcal{O}(x^7)$)	at Padé init.	–
Identity collapse possible	iff $\alpha = \lvert\beta\rvert$	Never

3.2. **Task-adaptive parameter trajectories.** A core advantage of Panh is its autonomous plasticity. Figure 5 displays the evolution of $\alpha(t)$ and $|\beta(t)|$ across structurally diverse PINNs tasks. The trajectories adapt dynamically based on the spectral properties of the target solution. While low-frequency problems maintain parameters near the Padé baseline, highly oscillatory targets dictate substantial parameter divergence to steepen non-saturating gradients. This confirms that Panh autonomously tunes itself without necessitating manual hyperparameter intervention.

3.3. **Identity-collapse monitoring.** Figure 6 plots the identity collapse monitor $\delta(t)$. Across all evaluated benchmarks, $\delta(t)$ remains bounded strictly away from zero. This empirically confirms that the Padé initialization successfully insulates the network against representation loss, ensuring the Panh layers actively retain nonlinearity throughout training.

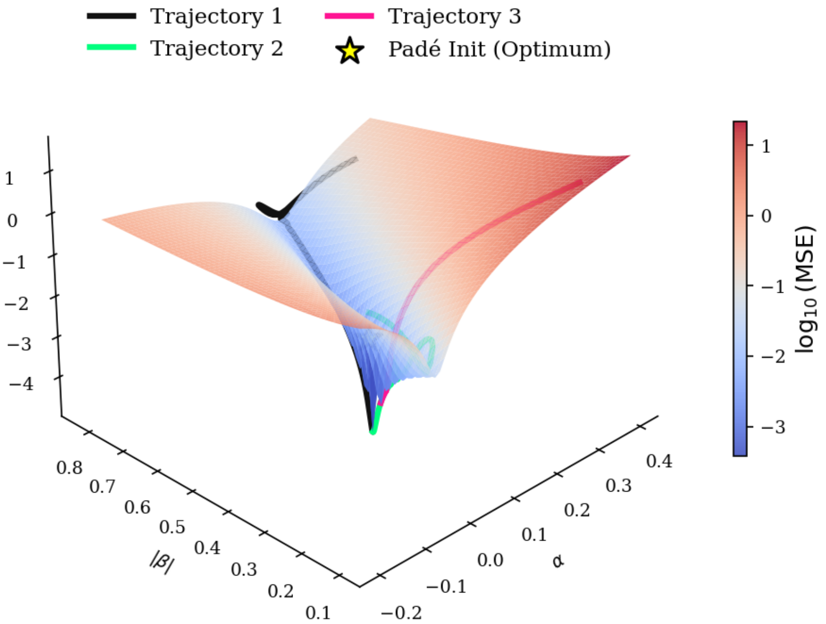

FIGURE 4. Loss landscape over $(\alpha, |\beta|)$. SGD trajectories from multiple random initializations natively descend into the optimal basin harboring the Padé point ($\lozenge$).

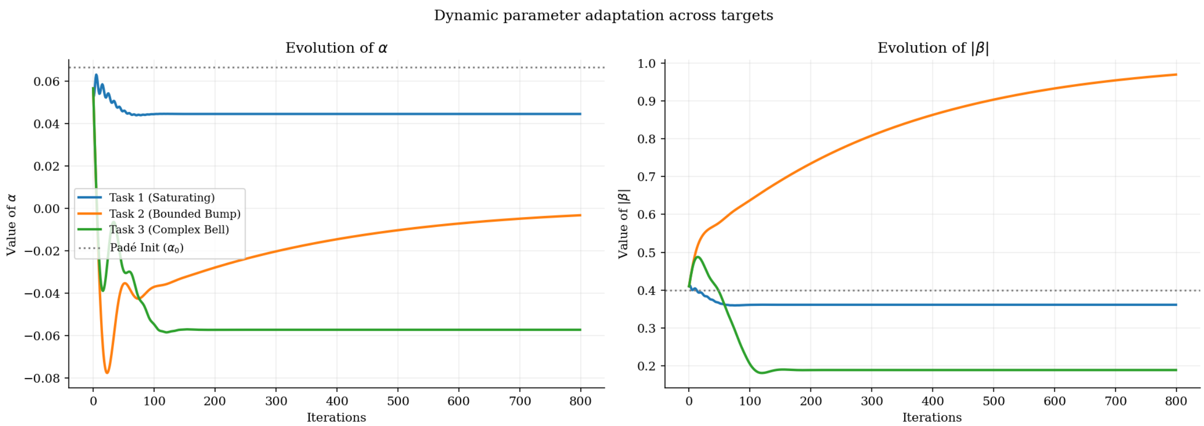

FIGURE 5. Evolution of adaptive parameters during training. Parameters converge to distinct problem-specific topologies depending on the physical task requirements.

4. Numerical Validation

We validate the theoretical claims across a suite of six benchmarks: three one-dimensional ordinary and partial differential equations, and three two-dimensional PDEs. This heterogeneous selection encompasses smooth unimodal targets, high-frequency oscillations, sharp rational peaks, and multi-scale superpositions.

4.1. **Experimental setup.** All experiments employ fully connected networks structured with four hidden layers of 40 neurons. The Panh architecture assigns an independently optimized parameter pair (α_i, β_i) to each layer, initialized at the Padé baseline. Models are optimized using Adam at a fixed learning rate of 10^{-3} for 3600 epochs. Model fidelity is measured via the relative L^2 error:

$$\varepsilon_{\text{rel}} := \frac{\|u_\theta - u^*\|_{L^2(\Omega)}}{\|u^*\|_{L^2(\Omega)}}, \tag{11}$$

where u_θ and u^* represent the network prediction and the exact solution, respectively.

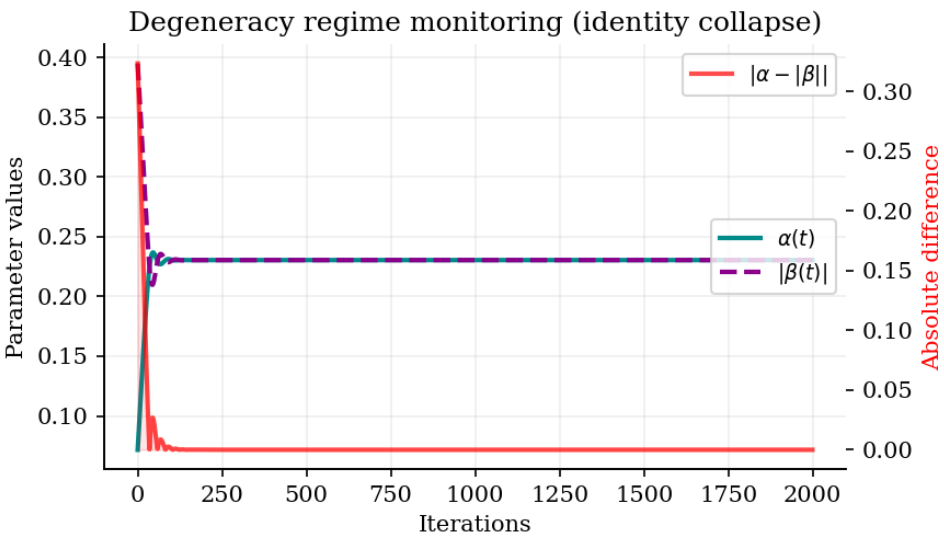

FIGURE 6. Identity-collapse monitor $\delta(t) = |\alpha(t) - |\beta(t)||$. The strict separation from the zero-bound confirms sustained non-linear expressivity.

4.2. **One-dimensional benchmarks.** The 1D evaluations address a high-frequency harmonic oscillator, the rational Runge function, and a multi-scale Poisson equation.

4.2.1. *High-frequency harmonic oscillator.* With $u = \cos(10x)$, the exact solution oscillates at a frequency substantially beyond the natural bandwidth of standard deep networks. As evidenced in Figure 7, the tanh network suffers from severe spectral bias, completely stagnating at $\varepsilon_{\text{rel}} = 5.41 \times 10^{-1}$. Conversely, Panh effortlessly resolves the oscillations, attaining $\varepsilon_{\text{rel}} = 1.74 \times 10^{-3}$, constituting an improvement factor exceeding 300. This is a direct empirical validation of Theorem 2.10.

4.2.2. *Runge function.* For the rational peak $u = 1/(1 + 25x^2)$, both activations converge successfully. Panh achieves 1.10×10^{-4} against 1.45×10^{-4} for Tanh. The performance parity here aligns with Theorem 2.6: near the origin, Panh accurately mimics Tanh, maintaining optimal representation for smooth bounded peaks.

4.2.3. *Multi-scale Poisson.* The superposition $u = \sin(\pi x) + 0.1 \sin(10\pi x)$ presents dual-scale spectral constraints. Tanh strictly fits the dominant low-frequency mode but ignores the high-frequency perturbation ($\varepsilon_{\text{rel}} = 3.50 \times 10^{-2}$). Panh resolves both elements precisely ($\varepsilon_{\text{rel}} = 4.64 \times 10^{-4}$), demonstrating accelerated and more robust convergence dynamics.

4.3. **Two-dimensional benchmarks.** The 2D tasks include a Gaussian peak, circular waves, and a nonlinear Poisson problem.

4.3.1. *Two-dimensional Gaussian peak.* For the localized peak $u = \exp(-20(x^2 + y^2))$, Panh reduces the maximum pointwise absolute error to 2.91×10^{-3}, compared to 4.55×10^{-3} for Tanh. As illustrated in Figure 8, the three-dimensional slice representation confirms that Panh reproduces the peak amplitude with superior fidelity.

4.3.2. *Two-dimensional circular waves.* The wave field $u = \sin(\pi(x^2 + y^2))$ presents an accelerating radial spatial frequency. Panh attains an error of 1.11×10^{-3} against 2.57×10^{-3} for Tanh (Figure 9). The error map confirms that Panh more evenly distributes residuals, preventing severe localized failures at higher frequencies.

4.3.3. *Two-dimensional nonlinear Poisson.* In this classic checkerboard domain $u = \sin(\pi x) \sin(\pi y)$, Panh achieves 1.14×10^{-3} compared to 3.34×10^{-3} for Tanh. Crucially, as observed in Figure 10, Tanh aggregates significant errors at the domain boundaries due to gradient saturation; Panh effectively suppresses this boundary-layer artifact, proving superior adherence to Dirichlet boundary constraints.

4.4. **Comparative summary.** Table 2 summarizes the performance metrics. Panh outperforms Tanh consistently across all configurations, highlighting its profound impact on high-frequency challenges and robust performance on well-conditioned problems.

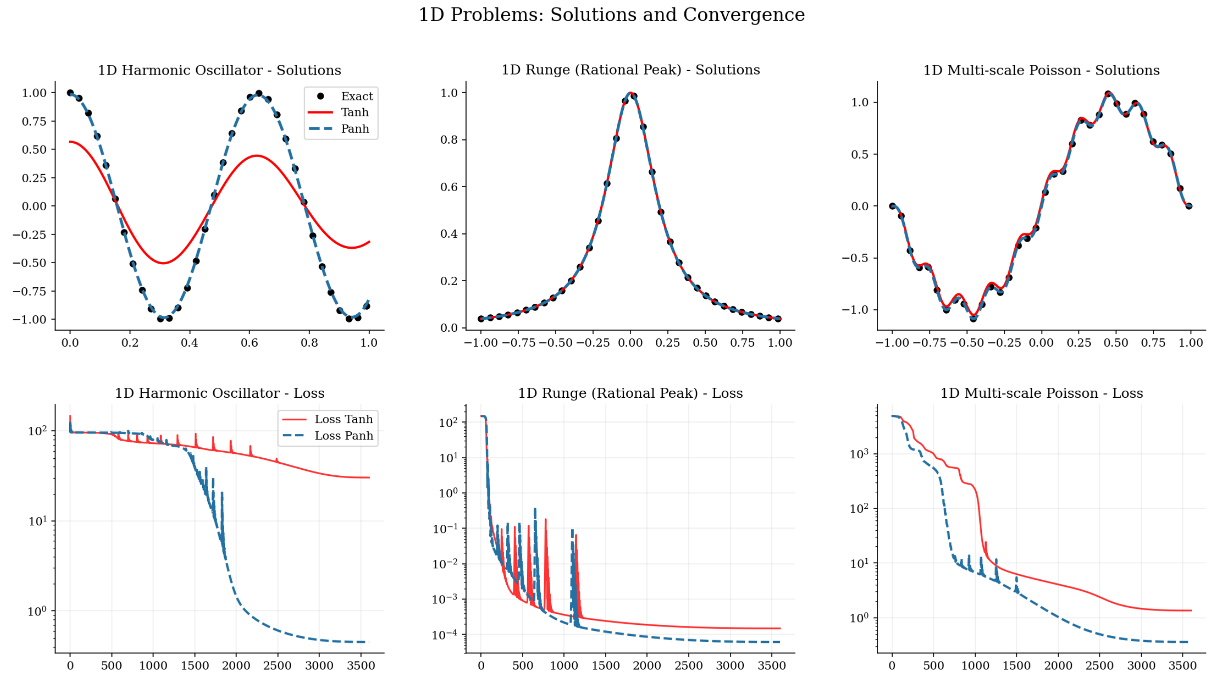

FIGURE 7. One-dimensional benchmark predictions and convergence histories. Panh accurately resolves high-frequency structures where Tanh experiences fatal spectral bias.

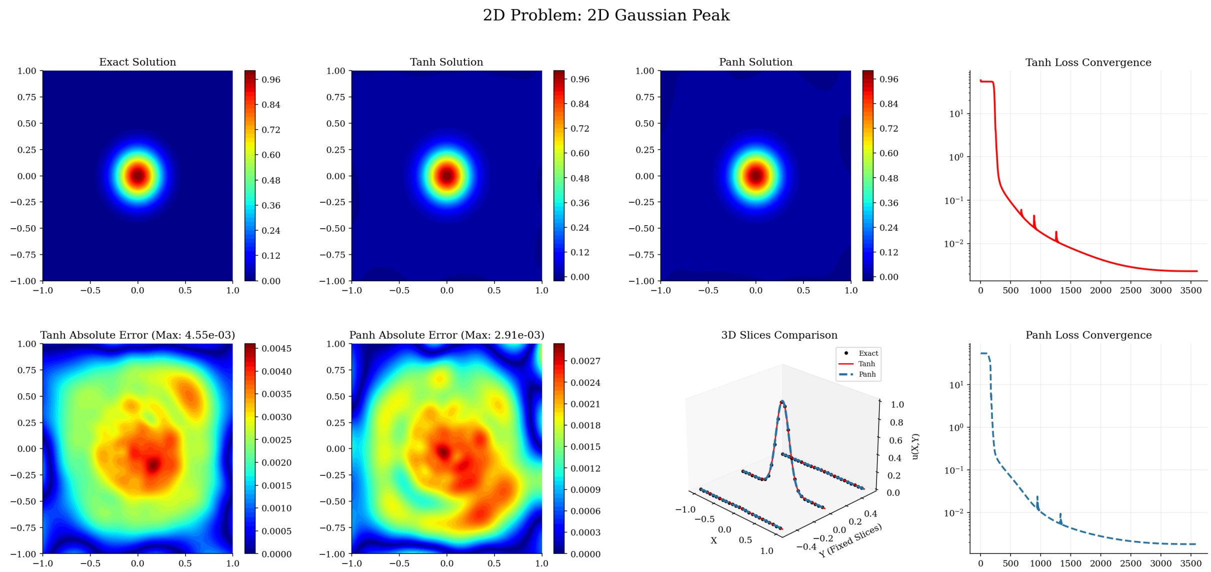

FIGURE 8. 2D Gaussian peak benchmark showcasing refined spatial accuracy and amplitude reproduction.

5. CONCLUSION

In this work, we introduced Panh, an adaptive rational activation function engineered to counteract the severe gradient pathologies inherent to standard saturating activations in physics-informed environments. By rigorously anchoring the formulation in Padé approximation theory, Panh guarantees a mathematically stable initialization while preserving the capacity to autonomously tailor its nonlinear geometry. Our explicit analytical derivations established its non-vanishing asymptotic gradient, factored its complex curvature, and derived a measurable condition indicating network collapse.

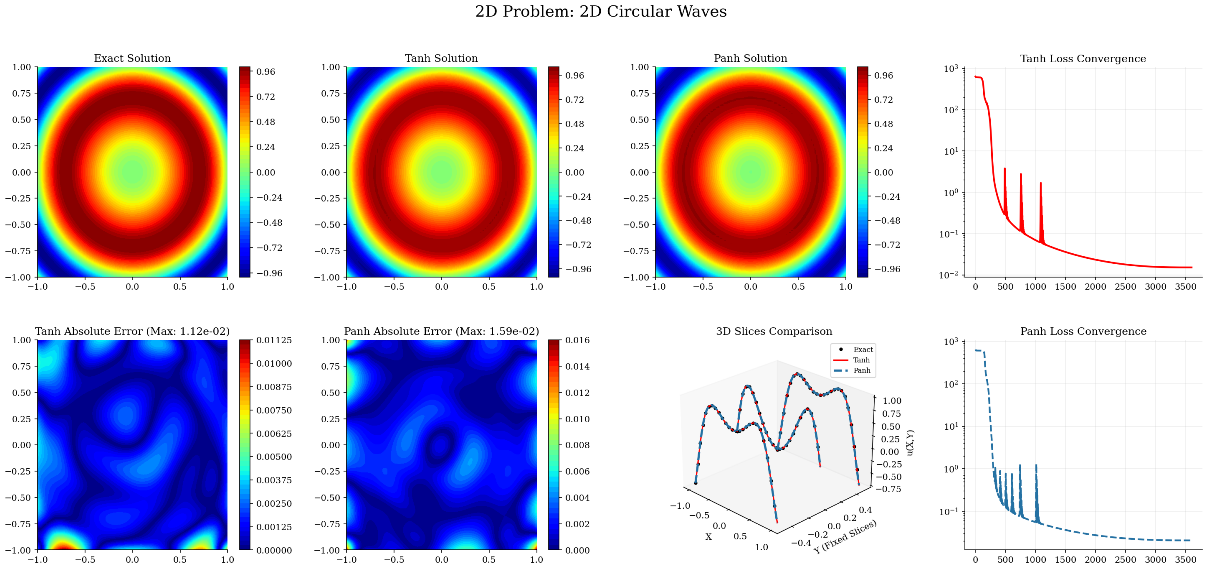

FIGURE 9. 2D Circular waves benchmark confirming superior radial frequency tracking capability.

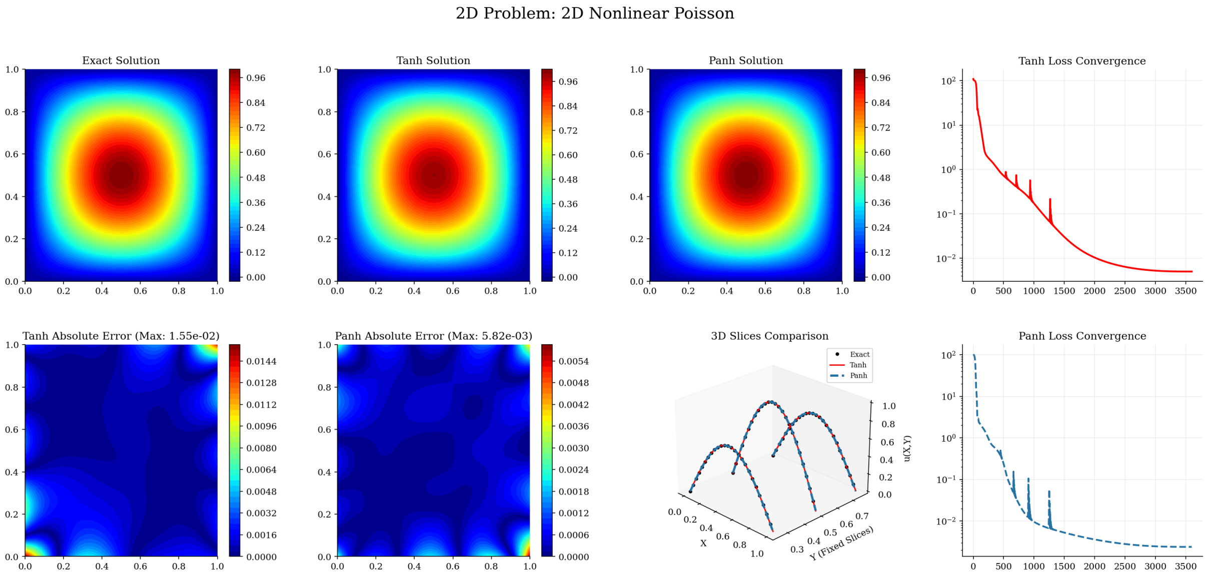

FIGURE 10. 2D Nonlinear Poisson benchmark. Panh eliminates the corner boundary-layer pathologies systematically observed in Tanh networks.

TABLE 2. Relative L^2 errors for evaluated benchmarks. The improvement factor underscores Panh's systematic advantage, particularly on stiff PDE instances.

Problem	$\varepsilon_{\text{Tanh}}$	$\varepsilon_{\text{Panh}}$	Improvement
1D Harmonic oscillator	5.41×10^{-1}	1.74×10^{-3}	**311.5**×
1D Runge function	1.45×10^{-4}	1.10×10^{-4}	1.3×
1D Multi-scale Poisson	3.50×10^{-2}	4.64×10^{-4}	**75.5**×
2D Gaussian peak	1.59×10^{-2}	1.08×10^{-2}	1.5×
2D Circular waves	2.57×10^{-3}	1.11×10^{-3}	2.3×
2D Nonlinear Poisson	3.34×10^{-3}	1.14×10^{-3}	2.9×

Comprehensive numerical evaluations confirmed that Panh dynamically alleviates spectral bias, substantially outperforming classical hyperbolic tangent activations. These performance enhancements surpassed two orders of magnitude in highly oscillatory domains. Future endeavors will evaluate the integration of Panh within advanced neural operator architectures, such as DeepONets, and rigorously analyze the spectral eigenvalues of its associated Neural Tangent Kernel.

Data and Code Availability

The source code and datasets required to reproduce the results generated during the current study will be made available by the corresponding author upon reasonable request.

References

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